

CS 370 Project Two

Design Defense

Jared Borck

6/21/26

The treasure hunt game is a pathfinding problem where the pirate intelligent agent needs to move through a maze and find the treasure. The environment includes open spaces, blocked areas, and a goal location. The purpose of the intelligent agent is to learn how to move through the maze successfully by using reinforcement learning. In this project, I used a deep Q-learning approach to train the pirate agent. The agent learns from experience by trying actions, receiving rewards or penalties, and updating its understanding of which moves are most useful. This allows the pirate to improve over time instead of being directly programmed with one fixed path.

A human and a machine would approach this maze in different ways. A human would likely start by looking at the layout of the maze, identifying walls and open paths, and trying to visually plan a route to the treasure. The human might choose a path that looks shortest, avoid obvious dead ends, and adjust the plan if the chosen route does not work. A person can also use memory and reasoning from previous maze experiences. For example, if a path leads to a blocked area, the human would remember that mistake and try another direction. This process uses visual understanding, prior knowledge, and logical reasoning.

The intelligent agent solves the problem in a more mathematical and experience-based way. The pirate does not understand the maze the same way a human does. Instead, it receives a state from the environment, chooses an action, observes the result, and receives a reward. Over many episodes, the agent learns which actions are better in different states. The agent is trained by repeatedly moving through the maze and updating its action values. Reinforcement learning is based on the idea that an agent learns through interaction with an environment and tries to maximize long-term reward (Sutton & Barto, 2018). In this project, the reward system helps the

agent learn that moving toward the treasure is good, while poor moves or invalid paths are less useful.

There are similarities between the human and machine approaches. Both approaches involve trial and error, learning from mistakes, and improving future decisions based on previous experience. A human may remember that a certain path did not work, while the intelligent agent updates values based on rewards and future expected rewards. The main difference is that humans can use general reasoning and visual judgment immediately, while the agent needs many training episodes to learn useful behavior. The machine approach is more repetitive and data-driven. It may take longer at first, but once trained, the agent can make decisions quickly and consistently.

The purpose of the pirate intelligent agent is to solve the pathfinding problem by learning a policy that leads it to the treasure. A policy is the agent's strategy for choosing actions in different states. In this game, the pirate needs to decide which move to make based on its current position and the maze layout. Reinforcement learning helps because the agent does not need to be given the exact solution path. Instead, it can learn from rewards. Q-learning is a model-free reinforcement learning method that estimates the expected value of actions and improves those estimates through experience (Watkins & Dayan, 1992). Deep Q-learning builds on this idea by using a neural network to approximate Q-values when the problem is too large or complex for a simple table.

Exploration and exploitation are important parts of this project. Exploration means the agent tries different actions to discover new paths and learn more about the environment. Exploitation means the agent uses what it already learned to choose the action that currently seems best. At the beginning of training, more exploration is useful because the agent does not know the best

path yet. If the agent only exploits too early, it may keep choosing a poor path and never discover a better one. Later in training, more exploitation is better because the agent has already learned useful information and should use that knowledge to reach the treasure efficiently. For this pathfinding problem, the ideal approach is to start with more exploration and gradually shift toward exploitation as the win rate improves.

In my implementation, deep Q-learning was used with a neural network to train the pirate agent. The neural network receives information about the current state of the maze and predicts Q-values for possible actions. The agent then chooses an action based on either exploration or exploitation. After the move, the experience is stored and used for training. The experience includes the current state, action, reward, next state, and whether the episode is complete. This process helps the model learn from previous actions instead of only learning from the most recent move. Deep Q-learning became an important method because it combines reinforcement learning with neural networks, allowing agents to learn action-value functions for more complex problems (Mnih et al., 2015).

The algorithm also used repeated training episodes so the agent could improve over time. During training, the agent attempted to move through the maze many times. When it made better decisions and reached the treasure more often, the win rate increased. The training process continued until the agent was able to complete the maze successfully. This shows how algorithms can solve complex problems by breaking them into states, actions, rewards, and updates. Instead of hardcoding every move, the algorithm learned a strategy through repeated interaction with the environment.

Deep Q-learning is useful for this game because it can handle a pathfinding problem where the agent must make a sequence of decisions. Each move affects the next state, so the agent has to consider more than just the immediate reward. It needs to learn which actions lead to better long-term outcomes. This is important because a move that looks useful right away might lead to a dead end later. The neural network helps estimate these long-term action values and supports better decision-making as training continues.

Overall, the pirate intelligent agent demonstrates how reinforcement learning can be used to solve a pathfinding problem. A human would solve the maze through visual reasoning, memory, and planning, while the machine learns through repeated actions and rewards. Both approaches involve learning from mistakes, but the intelligent agent depends on training data and reward feedback. The deep Q-learning algorithm allowed the pirate to learn a successful path to the treasure without being directly given the full solution. This makes it a strong approach for the treasure hunt game because the agent can improve through experience and eventually make effective decisions in the maze.

References

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